Complementary Threat Knowledge in Federated Android Malware Detection

A Controlled-Exposure Study

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Abstract

Collaborative training can help a malware detector recognize families it has never seen locally, but comparing federated with local training mixes this benefit with generic gains from pooling more data. We define complementary threat knowledge (CTK) as the recall gained when peers hold a client's hidden families, relative to training in which those families are absent from every client under the same budget. Using four Android market and era clients built from LAMDA and AndroZoo, leakage-aware partitions, per-client calibration, and ten confirmatory seeds, FedAvg CTK at a nominal 5% false-positive rate is 11.96 percentage points (95% BCa CI [9.15, 15.21]); the residual pooling effect, 1.51 points, has an interval spanning zero. Own-domain unseen-family recall rises 3.26 points over local training, while recall on nonhidden malware falls 3.61 points. Family-permutation and wrong-family placebo controls recover almost no gain, and CTK grows with peer dose.

Introduction

An Android malware detector can have many training samples yet none from a family of concern. Applications come from many sources, with different families, benign populations, and labels, so a detector will eventually face a family it never saw in training. Peers may hold examples of that family, but it is unclear whether collaborating with them improves detection of that family more than the additional data alone would.

Federated learning trains one model across clients by averaging locally learned parameters (McMahan et al., 2017). Comparing a federated model with a local one, however, confounds many factors: the number of clients, the training volume, and which families each client exposes. A federated model may lose overall performance yet gain the ability to detect families absent from a client, or the reverse.

We therefore define complementary threat knowledge (CTK). For a target client with a hidden family, CTK is the recall difference between federated training in which peers retain that family and federated training in which the family is absent from every client. The remaining difference between the family-absent federation and the local detector is called pooling. Together they decompose the total gain of collaboration over local learning, so a positive total can be attributed to complementary exposure or to generic pooling, and a negative pooling effect can be offset by complementary exposure.

We build four market and era clients from LAMDA features and AndroZoo metadata, hide selected families from target clients, and evaluate binary detection with per-client benign calibration. Group-aware partitions keep packages and identical first-level feature vectors from crossing the fit, calibration, and test splits, and every exposure condition uses the same training budget.

The chapter contributes (a) a matched-budget, three-arm design that separates complementary exposure from pooling in Android federated malware detection; (b) a confirmatory estimate of CTK over ten paired seeds, with negative controls, dose experiments, and robustness checks; and (c) a family- and client-level analysis of where collaboration helps and where it costs recall on known malware.

Background and Related Work

Android Malware Detection and Family Generalization

Static features from manifests and code, such as permissions and API calls, have long been used to classify Android applications and to explain which features mark an application as malicious (Arp et al., 2014). Because such classifiers learn whether an application is malicious rather than which family it belongs to, they can generalize to locally unseen families before any collaboration occurs. In CTK-Android, families are held out from training to control what each client can learn.

The representation determines how much shared structure can be found. MaMaDroid abstracts API calls to packages or families and models them as Markov chains (Mariconti et al., 2017). A family that shows no improvement under one representation may still share patterns that another representation can detect, so comparisons of CTK across representations must consider how the representation was extended, not only how many features were added.

Family labels are themselves uncertain. AVClass2 derives families from antivirus labels using predefined tags and rules (Sebastián & Caballero, 2020), and several datasets rely on such automated labeling. Families labeled this way may share an advertising library or otherwise overlap, so CTK-Android uses the provided annotations as given rather than as ground truth.

Federated Learning and Complementary Exposure

FedAvg aggregates locally trained models each round, and FedProx limits client drift from the aggregate with a proximal term (Li et al., 2020). Both serve as comparison points because their aggregation may compromise across clients when Android families are missing.

Ciaramella et al. (2026) compare federated and centralized malware detection using application-code images and convolutional networks under different data distributions and aggregation schemes. Gálvez et al. (2021) study privacy in federated malware classification, and Patel et al. (2026) address client-level concept drift. M2FD applies federated learning to mobile malware detection with drift monitoring on KronoDroid (Augello et al., 2025), and Droidware adds authenticated models and outlier checks to federated static-feature detection (Prasad & Kumar,

2026). These works inform deployment but do not isolate what a client gains from peers' exposure to families it lacks.

Similar questions arise in other security domains. Ongun et al. (2025) built CELEST, a federated method for detecting HTTP-based threats across networks, which measures performance when threat families are present in the federation but absent at a client, alongside active learning and defenses against malicious nodes. CyberForce varies how many IoT clients lack a malware type in federated reinforcement learning, including a setting in which every client lacks the target class, and compares aggregation algorithms (Feng et al., 2025). Shen et al. (2024) study intrusion detection when each client lacks one class, Popoola et al. (2022) examine missing unseen bots, and Shao et al. (2026) show that a model can converge yet remain biased when positives are missing. CTK-Android adapts these designs to Android by matching the number of replacement samples on every client, measuring recall across many hidden families with seed-level uncertainty, and adding wrong-family controls.

Dataset Quality and Evaluation Validity

AndroZoo provides millions of applications with metadata, allowing studies beyond small malware collections (Allix et al., 2016). We use its market field only as an observed attribute for constructing domains; retaining applications with a single market limits the corpus but makes domain membership unambiguous. A federation built from these domains is a benchmark, not evidence that market operators would behave like collaborating defenders.

Recent benchmarks extend Android data over longer periods to support drift analysis (Haque et al., 2026) and align several collection modalities for the same applications over time (Kamol et al., 2026). These resources inform our choice of data and representation but do not address complementary exposure.

Evaluation design can bias malware classifiers across time and space (Pendlebury et al., 2019), and common pitfalls include sampling bias and leakage between training and test data (Arp et al., 2022). CTK-Android addresses leakage from identical applications through grouped partitions.

It does not enforce temporal ordering within clients, a question Patel et al. (2026) study for Android federated learning.

Research Problem and Questions

A seed specifies a group-aware partition and selected families' target assignments. For learner m, $R(m, P)$ is a specified recall metric when targets lack their assigned families and peers retain them; $R(m, A)$ is the same metric when all selected hidden families are absent from every client's fit data. $R(L, P)$ is the recall of the local learner on the same target-family task, although peer data do not enter its training. All quantities share the seed, nominal operating point, and evaluation population:

$$\mathrm{CTK}(m) = R(m, P) - R(m, A)$$

$$\mathrm{Pooling}(m) = R(m, A) - R(L, P)$$

$$\mathrm{Total}(m) = R(m, P) - R(L, P) = \mathrm{CTK}(m) + \mathrm{Pooling}(m)$$

Each quantity is computed within a seed before averaging, and CTK and pooling can each be negative. Both federated arms use the same learner, training schedule, and per-client budget; withholding families changes only which replacement examples fill the budget. CTK therefore measures the combined effect of family exposure, replacement, and arm-specific calibration. Because several families are withheld in one run, a family's response can depend on the other exposures. Pooling also includes the difference in training schedule between federated and local learning, so it should not be attributed only to additional data.

The chapter addresses four questions. RQ1 asks whether peer exposure to a hidden family improves unseen-family recall once pooling is removed; the confirmatory endpoint is federation-wide client-macro unseen-family recall for FedAvg at a nominal 5% FPR. RQ2 asks how the benefit varies across families and target clients, and when it offsets negative pooling. RQ3 asks how learning strategies trade unseen-family recall against nonhidden-malware recall, achieved FPR,

and worst-client recall. RQ4 asks whether CTK persists under alternative family sets, support levels, partitions, model classes, aggregation rules, and representations. Full exposure, in which no family is withheld, serves as a finite reference for the same budget and learner; a better-sampled or longer-trained model could exceed it.

Controlled-Exposure Study Design

Data Sources and Constructed Clients

The source is the LAMDA var_thresh_0.01 release: 1,008,381 applications with 925 static feature columns. Each positive feature value is binarized, and no additional feature extraction or vocabulary construction is performed. Following the source, an application is malware if 4 or more antivirus engines detect it and benign if none do; applications in between are excluded. LAMDA covers 2013–2025 with no applications from 2015 (Haque et al., 2026); we construct our own splits.

All 1,008,381 rows were linked by SHA-256 to exactly one AndroZoo metadata record with a matching VirusTotal count. Only applications associated with a single market are retained. Anzhi and AppChina form one client each, and Google Play applications are split into play-early (before June 2019) and play-late (after June 2019); other applications are not used. Table 1 shows the resulting 832,165 applications: 257,564 malware and 574,601 benign.

Table 1
Retained Applications in the Four Constructed Clients

Client	Applications	Malware	Benign	Malware (%)
anzhi	104,287	90,801	13,486	87.07
appchina	88,282	73,578	14,704	83.34
play-early	319,122	75,985	243,137	23.81
play-late	320,474	17,200	303,274	5.37

Note. Corpus counts before grouped splitting and fit-budget selection. Client labels denote constructed market/era domains; percentages describe this selected corpus.

Class proportions differ sharply across clients (Table 1): Anzhi and AppChina are mostly malware, and both Google Play clients are mostly benign. These proportions are retained rather

than balanced, so an equal recall gain may affect each client differently. They describe the selected corpus, not market prevalence.

Each corpus is larger than the training budget. In the main setting each client fits at most 6,000 examples per arm (1,500 in the lower-support sensitivity), while calibration and test sets keep their full partition groups. Training volume is therefore identical across arms, and available peer support can exceed what is fit.

Identity Grouping and Partition Construction

Two applications are linked if they share a package name or an identical binary feature vector. Connected components of this relation, including transitive links, are atomic: each component receives one role for the entire study, even when it spans several clients. No package or primary feature vector therefore appears in more than one of the fit, calibration, and test sets, while each set still contains every client.

Components are ordered by seed, partition salt, and attempt number and assigned cumulatively to fit, calibration, and test at target shares of 60%, 20%, and 20%; because components are atomic, realized row shares can differ. For each seed and salt, 16 attempts are generated and the one with the most eligible fit and natural-scarcity client families is kept, with ties going to the first attempt. Selection uses support only, not model performance. Across confirmatory seeds the selected partitions have 57–60 eligible client families, against candidate medians of 52.5–56, about 9% more support; results are conditional on this selection.

Partitions are not temporal: both Google Play clients have their own fit, calibration, and test sets, so an application's date does not guarantee that it was fit before it was tested. Generalization is therefore to new identities within each market or era. A package-only grouping, which drops feature-vector links, serves as a sensitivity analysis; it does not rule out other forms of leakage.

Family Selection and Target Assignment

Empty, unknown, benign, and singleton family labels are excluded as targets, although those applications remain in binary learning. Families with at least 600 applications in the retained corpus and presence in at least 2 clients are candidates. Candidates are sorted by support, and the top 8 for

each set are assigned deterministically by alternating ranks between a primary and a replication set. The primary set contains 8 names, including frmy, of which 7 meet the controlled-study requirements; the replication set contains 8.

A target–family pair is eligible when, before budget selection, the target has at least 20 fit examples of the family to withhold, peers have at least 150 fit examples, the federation test set has at least 50 examples, and at least 500 target fit examples remain after removal (summed over all families assigned to that target). Clients are ordered by seed and the first eligible client becomes the target, so a family's target can change across seeds.

Own-domain reporting additionally requires at least 15 hidden-family test examples on the target's own domain. A family can therefore contribute to federation-wide results but not to own-domain results, and the number of contributing clients and seeds differs between the two populations.

Exposure Conditions and Comparison Designs

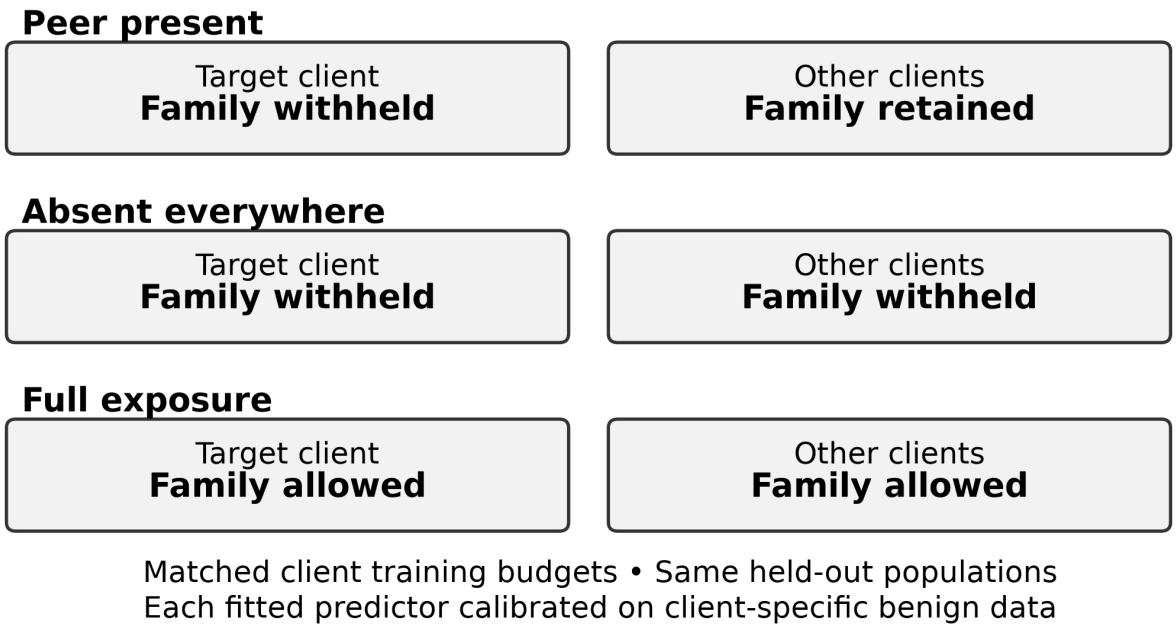
Figure 1 shows the three conditions. In peer-present, each selected family is withheld only from its assigned target; in absent-everywhere, all selected families are withheld from every client; full exposure removes the deliberate exclusions. After exclusion, seeded ordering decides which rows fill each client's budget, so sample counts are identical while composition differs.

Replacing withheld examples keeps each client's sample count constant but changes its malware–benign mix, and each arm is recalibrated on its own benign scores. CTK therefore captures the combined exposure, replacement, and recalibration effect of the selected families, not the contribution of one family in isolation.

Natural scarcity is a sensitivity analysis in which a family's target holds less than 5% of that family's federated fit, with at least 150 peer fit examples and 15 target test examples. The target family is not removed, so a few local training examples may remain, and the selected families and scarcity definition differ from the controlled design; it is therefore a sensitivity analysis rather than an equivalent estimate.

Figure 1

Controlled Family Exposure Across Target and Peer Clients



Note. The schematic illustrates one target-family pair. Each run can assign several families to different targets. In the peer-present condition, a selected family is withheld only from its assigned target, which may retain other selected families. In the absent-everywhere condition, all selected families are withheld from every client. Replacement examples preserve the 6,000-sample training budget per client. Full exposure removes the deliberate exclusions but retains the budget.

Aligned Representation Extension

Intersecting the retained LAMDA applications with McNdroid by application hash leaves 554,136 applications (170,256 malware and 383,880 benign). Four representations are compared on this shared population: 925 LAMDA static features, 2,390 McNdroid static features, 2,793 call-graph columns, and 17,483 report-JSON columns. The graph features are those McNdroid derives; no new graph model is built. Report features are sign-log transformed. Call-graph and report columns are kept when at least 1% of the training rows are positive and are standardized with training statistics; each trained model fits these transformations on its own training rows only (a client's rows for local models, the arm's pooled rows for central and FedAvg models).

Every representation keeps the same labels, clients, and family names from the shared population. Family-level comparisons focus on hiddad, gappusin, revmob, leadbolt, airpush, dowgin, and adwo, all of which were studied earlier, so these are internal rather than external comparisons.

Learning and Evaluation Protocol

Learners and Frozen Training

Table 2 lists the frozen settings. The base model is an MLP trained with binary cross-entropy on logits. Local models train for 10 epochs on their client's budgeted fit sample, and the central model trains for 20 epochs on all four budgeted samples. FedAvg runs 20 rounds in which all 4 clients start from the current global parameters, train 2 local epochs with a newly created Adam optimizer, and are averaged with weights proportional to fit examples (equal here, because every client fits 6,000). FedProx adds a proximal term; fine-tuned FedAvg trains the final FedAvg model for 2 more epochs on the target's fit data; and the blend averages local and FedAvg predictions with equal weight. Every predictor is calibrated on its client's benign calibration scores. Local epochs, fine-tuning length, and proximal strength were chosen on a development set rather than on test results, and they do not guarantee that each learner reaches its optimum.

Two alternative model classes test whether the comparison depends on the MLP: a linear classifier, trained locally, centrally, and with FedAvg, and a histogram gradient-boosted tree learner with at most 100 estimators and depth 6, trained locally and centrally.

Calibration and Metric Definitions

For each client and arm, the threshold is the empirical $(1 - \alpha)$ quantile of the benign calibration scores, taking the higher order statistic, and an application is flagged when its score exceeds the threshold. The primary operating point is $\alpha = .05$, with sensitivity analyses at .01 and .10. Calibration is inadequate when $n_{\text{benign}} \times \alpha < 10$. Because of score ties and differences between calibration and test populations, achieved test FPR varies around α; at the primary operating point the run-level client-macro FPR must lie within ±2 percentage points of α. This check does not bound each client's FPR.

Table 2
Conditions of the Primary Controlled Study

Component	Setting
Source and representation	LAMDA var_thresh_0.01; 925 binary features
Partition	Package/feature connected components; 60/20/20 fit/calibration/test; best support among 16 attempts
Training volume	6,000 fit rows per client; 1,500 in lower-support sensitivity
Primary eligibility	Target fit ≥20; peer fit ≥150; federation test ≥50; remaining target fit ≥500
MLP and optimization	256/64 hidden units; rectified linear activation; dropout 0.1; Adam; learning rate 0.001; batch 256
Training schedule	Local 10 epochs; central 20; federated 20 rounds × 2 local epochs
Adaptation	FedProx $\mu = 0.1$; fine-tuning 2 epochs; equal probability blend
Operating points	Nominal FPR 5% primary; 1% and 10% sensitivity
Primary inference	Seeds 100–109; paired effects; 9,999 BCa resamples; exact Wilcoxon; Holm across three contrasts

Note. Support requirements refer to pre-budget pools. Family-macro, own-domain, and worst-client summaries use distinct denominators. Hyperparameters and statistical procedures were frozen before confirmatory evaluation. The stored resolved configurations of all confirmatory runs match the frozen run-level protocol.

Own-domain unseen-family recall is the share of a target's own-domain test applications from its hidden families that are flagged, averaged over targets meeting the own-domain support requirement. Federation-wide unseen-family recall applies the target's detector to its hidden families in all four domains and averages over targets; it has more support but includes applications outside the target's market or era. Family-macro recall averages recall over eligible target families with equal weight, whereas micro-pooled recall divides total hits by total trials and therefore weights high-support families more. The confirmatory estimand is federation-wide client-macro recall; family-macro recall is used in the dose and family analyses, and estimates can differ with the weighting.

Nonhidden-malware (known-family) recall is the client-macro recall on test malware outside each client's hidden families, which may include unlabeled families. Achieved FPR is the client mean on benign test applications, and worst-client FPR is the largest client FPR. Worst-client unseen recall is the smallest federation-wide hidden-family recall among targets within a seed; because the worst client can differ between arms, its difference is not one client's improvement.

AUROC and average precision (scikit-learn average_precision_score) are computed per client over all own-domain test applications and averaged equally. Average precision sums precision at each recall increment rather than interpolating the precision–recall curve with trapezoids, and it depends on class proportions. Neither metric is restricted to hidden families or to the operating point.

Recall differences are reported in percentage points, and recall and FPR in percentages. Nominal FPR is the rate targeted on calibration data; achieved FPR is measured on benign test applications. Intervals are 95% BCa intervals for means unless stated otherwise.

Statistical Inference and Evidence Status

Confirmatory seeds are 100–109. Within each seed, arms share the partition and targets, so differences are paired. We report the mean, median, number of positive seeds, and standardized effect size, and form 95% BCa intervals from 9,999 bootstrap resamples of seed differences. With no zero or tied absolute differences among the three primary contrasts, exact two-sided Wilcoxon signed-rank tests assess whether differences are symmetric about zero, not whether the mean is zero. Holm correction (Holm, 1979) at $\alpha = .05$ covers only the three FedAvg federation-wide contrasts (CTK, pooling, and total); other learners, populations, family analyses, and diagnostics are unadjusted.

Practical thresholds were fixed before evaluation and are not significance levels: CTK of 3 percentage points is considered useful, and nonhidden-malware recall may fall by at most 2 points. Bootstrap intervals and signed-rank tests answer different questions, and neither method was changed when they disagreed.

Evidence is labeled by class: A, original confirmatory; B, prespecified sensitivity; C, post-confirmatory analysis of saved outputs; and D, separately frozen prospective extensions and post hoc diagnostics. Extensions use fresh seeds: 200–209 for permutation, 300–309 for the expanded family set, 310–319 for exact dose, 320–329 for placebo and aggregation, and 330–339 for representation. They are never combined with seeds 100–109 as confirmatory evidence.

Code, configurations, outputs, and data fingerprints are available at https://github.com/naslo uby-salahe/ctk-android (run revision f6302f7906824945731dadbbeb44357f3d025aaf). Reported results come from the final deterministic run, with seeded GPU dropout and deterministic kernels. Two earlier runs with the same settings were not reproducible bit for bit and gave FedAvg CTK of ~11.7 points; they are not pooled with the reported run. Source data must be obtained separately.

Experimental Results

Primary Controlled-Exposure Results

Peer exposure produced a clear complementary gain (Table 3). FedAvg federation-wide unseen-family recall rose from 52.41% with the selected families absent everywhere to 64.36% when peers held them; local learning reached 50.90% and full exposure 67.64%. Mean achieved FPR was 5.04% for absent-everywhere, 5.09% for peer-present, and 5.04% for full exposure, so the arms are compared at similar operating points. The mean hides client variation: mean worst-client FPR was 6.60% for peer-present, with a maximum of 9.89% across seeds.

Table 3
Primary Paired FedAvg Decomposition at Nominal 5% FPR

Contrast	Mean (pp)	95% BCa CI (pp)	Holm *p*	Positive seeds
CTK	+11.96	[+9.15, +15.21]	0.00586	10/10
Pooling	+1.51	[−4.15, +6.10]	0.32227	7/10
Total	+13.46	[+8.82, +17.04]	0.00586	10/10

Note. Ten confirmatory seeds, 100–109. CTK = peer present − absent everywhere; pooling = absent everywhere − local; total = peer present − local. Realized FPR: local 4.99%, absent 5.04%, peer 5.09%, full exposure 5.04%. Intervals summarize seed-paired mean differences; *p* values are exact two-sided signed-rank results adjusted across these three contrasts. pp = percentage points.

CTK and total gain were positive in all 10 seeds, while the pooling interval included both small negative and small positive effects. On average 2,073 of 24,000 fitted samples differed between the peer-present and absent-everywhere arms.

On own-domain unseen malware, FedAvg CTK was 10.85 points (6.97–15.25), pooling was −7.60 (−11.81, −4.03), and the total gain over local learning was 3.26 (1.26, 5.63). Most own-domain

CTK therefore repairs the loss from pooling. These own-domain estimates are secondary and not Holm adjusted.

Other aggregations agree in sign. Family-macro CTK was 12.44 (10.09–15.55) with a total gain of 10.57, and worst-client CTK was 18.49 (13.88–26.29) with a total of 14.88. Weighting each hidden test application equally (micro-pooling) gave a smaller CTK of 6.58 (4.64–8.65), still positive in all 10 seeds, with pooling of 4.00 and a total of 10.58. The difference comes from dowgin, which supplies about 47% of hidden-family test applications and shows little CTK; removing it raises micro-pooled CTK to 12.24 points. At the alternative operating points, FedAvg federation-wide CTK was 7.85 points at 1% nominal FPR and 13.78 at 10%.

Learning-Strategy Comparisons

Table 4 compares learners in the peer-present condition. Every alternative learner showed positive CTK in all ten seeds: 13.10 points for central training, 13.20 for FedProx, 8.20 for fine-tuned FedAvg, and 7.60 for the blend. FedProx minus FedAvg federation-wide unseen recall was 0.19 points (−1.12, 2.94), too small to rank the two. Central training had the highest descriptive federation-wide mean, and FedAvg the highest mean worst-client recall.

Table 4
Peer-Present Learner Performance at Nominal 5% FPR

Learner	Federation unseen	Own-domain unseen	Worst client	Nonhidden	FPR
Local	50.90	50.67	28.92	73.06	4.99
Central	66.07	56.27	41.42	72.92	5.10
FedAvg	64.36	53.93	43.79	69.46	5.09
FedProx	64.55	55.98	37.90	72.13	5.09
Fine-tuned FedAvg	62.16	51.64	40.35	69.33	5.03
Blend	63.66	57.21	37.63	74.77	5.12

Note. All entries are percentages averaged over ten confirmatory seeds. Worst client is the within-seed minimum federation-wide unseen recall. Other recall and FPR columns follow the macro definitions in the protocol. Arm rankings are descriptive; table levels are not confidence intervals or adjusted pairwise superiority tests.

Nonhidden-malware recall changes the ranking. Relative to local learning, FedAvg lost 3.61 points (−4.50, −2.85) and fine-tuned FedAvg 3.74 (−5.31, −2.82), both beyond the 2-point tolerance;

FedProx lost 0.93 and central training 0.14, and the blend gained 1.70. These compare means with a fixed tolerance and are not equivalence tests. FedProx and the blend retained more nonhidden recall but had lower worst-client recall than FedAvg.

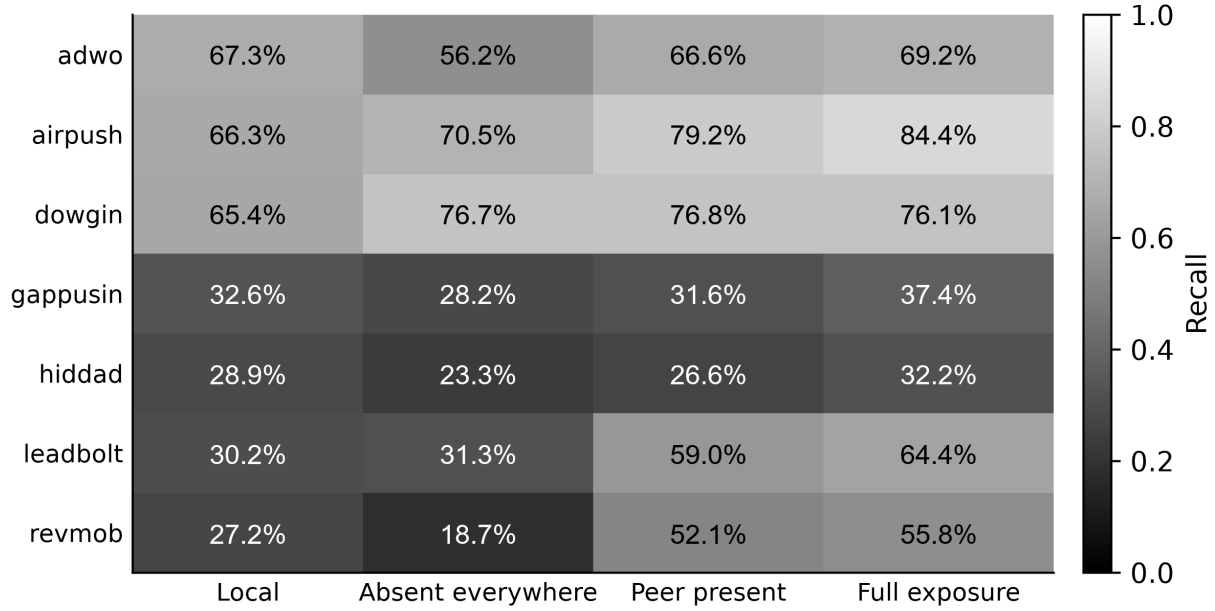
Gains on hidden families came with a small loss in overall discrimination: FedAvg minus local changed mean own-domain AUROC by −0.01166 and average precision by −0.02011. Both views are needed to evaluate a detector.

Malware-Family and Client Heterogeneity

Figure 2 shows primary-family recall by condition. Revmob and leadbolt gained most, with CTK of 33.38 and 27.75 points, yet revmob stayed below 60% recall even under full exposure, so a large gain need not be sufficient.

Figure 2

Family-Specific Recall Across Exposure Conditions



Note. Cells show mean recall percentages to one decimal place at nominal 5% FPR across seeds 100–109. Full exposure refers to FedAvg and is a finite reference. The grayscale bar expresses the same recall values as proportions.

Families responded in different ways. Dowgin gained mainly from pooling: its recall rose well above local even when it was absent everywhere, and its CTK was only 0.14 points. Adwo showed repair of negative transfer: its CTK of 10.37 points almost offset the loss in the absent-everywhere

arm, although peer-present recall remained below local. Gappusin and hiddad responded weakly to the static representation, with CTK of 3.46 and 3.25 points but peer-present recall of 31.63% and 26.57%, below their local recall; full-exposure FedAvg recall reached only 37.41% and 32.18%. These two families raise a question about representation more than about exposure. Airpush was intermediate, with CTK of 8.71 points and peer-present recall of 79.24%.

In the replication set, CTK was negative for dnotua (−8.24 points) and zdtad (−2.06) and positive for inmobi (13.63) and domob (12.04); utchi and zdtad contributed only six seeds. The replication aggregate was positive, but not every family benefits from exposure.

By client (class C), federation-wide FedAvg CTK was 8.35 points at Anzhi, 7.58 at AppChina, 9.18 at play-early, and 32.03 at play-late, from 8, 10, 10, and 6 contributing seeds; play-late's high value rests on fewer seeds and about 107 own-domain hidden test applications per seed. Anzhi's total gain was only 1.34 points despite positive CTK, and its nonhidden-malware recall fell 6.93 points. Own-domain totals were −2.80 at Anzhi, −0.40 at AppChina, 6.48 at play-early, and 13.61 at play-late, with intervals crossing zero for the first two, so the aggregate own-domain gain is not uniform across clients.

Peer Exposure and Dose Response

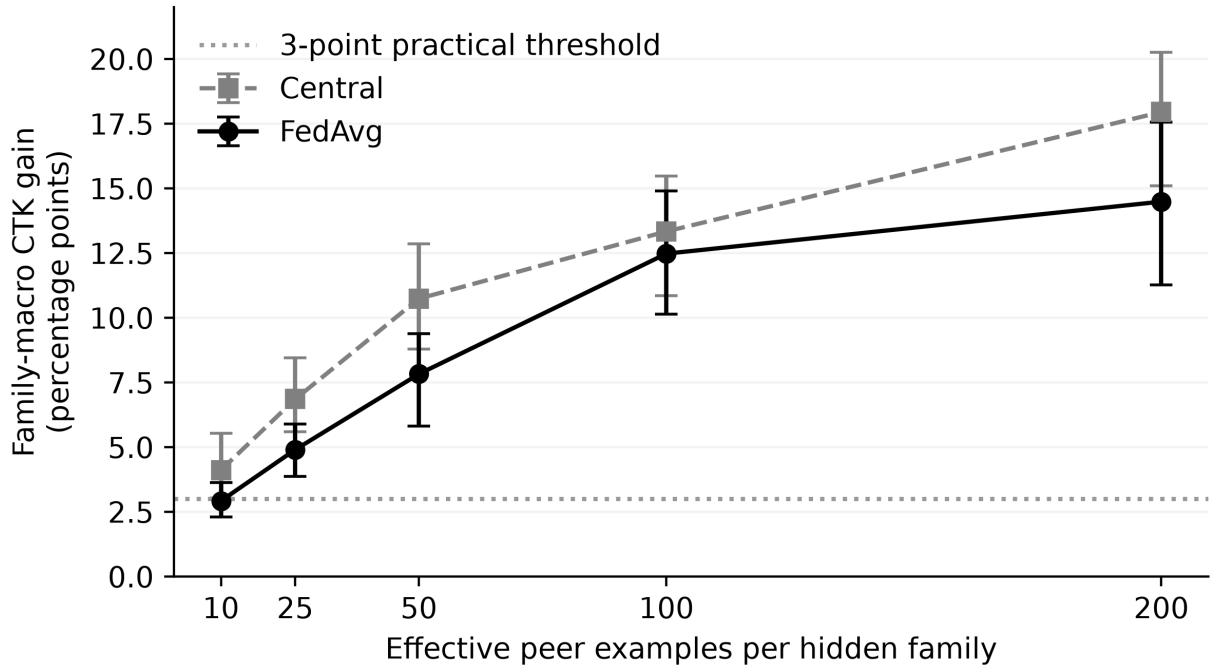
Requested peer caps overstated actual exposure. With caps of 100 and 1,000, the mean effective dose for FedAvg was 9.03 and 79.39 fitted peer examples, because peers were sampled within budgets after capping; with all available examples it was 296.19, with family-macro CTK of 12.44 points. Only the all-available condition exceeded 100 fitted peer examples per family on average.

In the exact-dose extension, pooled family-macro federation-wide FedAvg CTK was 2.92, 4.90, 7.83, 12.48, and 14.49 points at 10, 25, 50, 100, and 200 effective peer examples (Figure 3), positive in all 10 seeds at every dose. At 10 examples the interval (2.31–3.64) contains the 3-point threshold; at 25 it is 3.87–5.89.

The curve rises without a clear saturation point. Pooled CTK increased 4.65 points from 50 to 100 (2.43 to 7.09) and 2.01 from 100 to 200 (−0.06 to 4.06). From 100 to 200 the primary set

Figure 3

Exact Peer Dose and Complementary Gain



Note. Prospective extension, seeds 310–319. Family-macro federation-wide effects pooled within seed across primary and replication sets at nominal 5% FPR; 95% BCa intervals. The dotted line marks a practical threshold, not a universal dose requirement.

still gained 4.23 points (2.53 to 6.55), whereas the replication set was flat at 0.03 (−3.91 to 2.54). Families share no saturation point or minimum dose, and dowgin stayed near or slightly below zero even at the highest dose. By construction, the zero-dose arm equals absent-everywhere.

Negative Controls, Replication, and Robustness

The family-permutation control breaks the alignment between family labels and applications while preserving malware status. Original FedAvg federation-wide CTK under permutation was 0.30 points (−0.36 to 1.02), equivalent to zero under the ±3-point criterion; the own-domain interval reached 3.04 and remained unresolved. In the extension, federation-wide and own-domain CTK were −0.21 and −0.07 points, both within ±3 points.

Table 5 summarizes replication, sensitivity, and control effects. The wrong-family placebo, which reallocates support from a different family, produced −0.54 points, and CTK minus placebo

was positive in all ten extension seeds, indicating that the gain is specific to the correct family rather than to any reallocation of support.

Table 5
Selected Replication, Sensitivity, and Control Effects

Comparison	Class	Mean (pp)	95% BCa CI (pp)
Disjoint family set	A	+5.08	[+2.28, +7.10]
Natural scarcity	B	+11.19	[+9.46, +12.59]
Linear FedAvg	B	+9.37	[+7.54, +10.97]
Centralized trees	B	+10.18	[+7.59, +13.70]
Lower training support	B	+10.99	[+8.71, +13.52]
Lower family support	B	+14.01	[+11.46, +16.22]
Higher family support	B	+11.59	[+9.17, +13.75]
Package-only grouping	B	+11.32	[+8.60, +13.29]
Wrong-family placebo	D	−0.54	[−1.36, +0.20]
CTK minus placebo	D	+10.04	[+8.35, +12.28]

Note. A = original confirmatory; B = prespecified sensitivity; D = prospective extension. First eight rows are CTK effects on the client-macro federation-wide metric, seeds 100–109. Last two are family-macro effects pooled within seeds across family sets, seeds 320–329. These rows are not one multiple-testing family and are not pooled into the primary test. A positive mean above three points need not have an interval entirely above that practical threshold.

The disjoint replication family set was positive in 8 of 10 seeds with mean CTK of 5.08 points, but its interval does not lie entirely above the 3-point threshold. Linear FedAvg, central trees, lower training support, lower and higher family support, package-only grouping, and three alternative partition salts all gave positive mean CTK, as did natural scarcity, although natural scarcity also differs in local exposure, targets, and pooling. These checks extend the result within the same corpus and construction.

With robust aggregation, pooled CTK was 9.50 points under mean aggregation, 9.16 under a coordinate-wise trimmed mean that drops the highest and lowest of the four client values, and 13.76 under a coordinate-wise median that takes the lower middle value. These analyses assume honest clients and do not test defenses against malicious updates.

Representation and Expanded-Family Analyses

The aligned-representation extension separates gains in full-exposure recall from gains in CTK. On the priority-family macro, McNdroid static features increased central full-exposure recall by 7.57 points (2.76–12.85) and FedAvg CTK by 8.01 (3.56–10.54) relative to aligned LAMDA. Call-graph features increased CTK by 9.03 (5.53–12.35) but full-exposure recall by only 2.52 (−0.86 to 7.77), so a higher CTK does not imply higher full-exposure recall; CTK also depends on recall when families are absent. Report-JSON features lowered full-exposure recall by 14.54 points (−18.63 to −8.66) and CTK by 6.12 (−9.91 to −3.87). Family responses varied: static features raised full-exposure recall for gappusin by 16.93 points and adwo by 33.60, but hiddad's estimate of −6.21 (−18.02 to 6.29) was inconclusive.

These comparisons use nominal calibrated thresholds, not equal achieved FPRs. Macro-FPR tolerance failed for exact-dose primary seed 317, LAMDA-static and McNdroid-static representation seed 333, and call-graph seed 334, where one central peer-present client reached 15.32% FPR. A post hoc diagnostic that equalizes test FPR using benign test scores gave similar full-exposure effects (+7.28 points for static features and −14.27 for report JSON), while the call-graph full-exposure interval contained zero. This diagnostic is not a deployable procedure.

The expanded set of 32 families had a mean family CTK of 6.51 points; 5 families were negative, including feiwo (−5.89) and baiduprotect (−4.41). Distance from the nearest well-supported known family correlated with CTK (Spearman correlation .408, family-bootstrap interval .088–.662, permutation $p = .0214$), but among the 27 families eligible in all fresh seeds the correlation weakened to .294 (−.076 to .589, $p = .136$). This distance is not a direct measure of novelty and cannot predict which partners will help. Of the 32 families, 14 overlapped the primary or replication sets; the 18 new families averaged 3.47 points of family-macro CTK per seed (2.45–4.45), positive in all fresh seeds, against a mean of 10.67 for the overlapping families. This result is supportive but is not a second confirmation.

Discussion and Practical Implications

What the Controlled Contrast Establishes

The matched-budget contrast in Figure 1 estimates the combined effect of selected-family exposure, replacement samples, and arm-specific calibration while holding training time and client samples fixed, which a local-versus-federated comparison cannot do. Most of the federation-wide gain comes from complementary exposure rather than generic pooling. On a client's own domain, pooling is negative and CTK mainly repairs that loss, so a client should ask how collaboration compares with its own detector and why. Families differ in mechanism: adwo recovers a pooling loss, revmob and leadbolt exceed local detection, dowgin benefits without exposure, and hiddad remains hard. Disjoint and newly identified families also benefit, but less. These patterns may change with other corpora or features.

Evaluating Collaboration as a Set of Tradeoffs

The evaluation population matters when choosing an algorithm. Federation-wide recall asks whether a family is detected across source domains; own-domain recall asks whether a client's own applications benefit. Similarly, family-macro recall weights families equally but does not count how many attacks would be stopped. Learner choice involves hidden-family recall, nonhidden-malware recall, achieved FPR, and worst-client recall together: FedAvg had the best mean worst-client recall but lost the most nonhidden recall, FedProx limited that loss without improving mean unseen recall, and fine-tuning and blending shifted the balance further. Operators must decide which losses they accept and test on their own populations.

FPR also needs attention. The mean is close to nominal, but some clients and extension seeds exceed the tolerance, and a 5% benign FPR may be too high for triage, which this study does not measure. Aggregate metrics such as AUROC and average precision can hide losses on hidden families, and hidden-family recall can hide losses elsewhere, so reports should state the population, threshold, and client averaging.

Exposure, Representation, and Remaining Questions

Peer support should be measured after exclusion and budget selection, because nominal caps overstate exposure. The exact-dose experiment tracks exposure precisely but does not tell how many examples a new domain or representation will need. Representation matters as well: some families benefit from richer static information and others do not, and a higher CTK need not mean higher full-exposure recall. Because the aligned data overlap the main corpus, these conclusions hold only within the same family-label system.

Privacy is not addressed: clients were not protected and no privacy budget was applied, and positive CTK under trimmed or median aggregation shows only that the gain survives in an unpoisoned federation. Centroid distance conflates support, representation, and eligibility, so controlled exposure is a better guide to partner value than correlation with distance. Testing on unrelated domains, previously unseen families, and temporal splits remains necessary before production use.

Threats to Validity

Construct and Internal Validity

CTK is measured against the provided family annotations, which may merge or split behaviors, and the selected families lean toward adware rather than covering all Android threats. Locally unseen means absent from the fit set, not unknown in general. A weak response can reflect labeling, low support, poor representation, or the fitted rule itself, and the design cannot separate these causes. Several families were withheld at once, so arms may interact through replacement examples. Grouping removes exact package and first-level feature matches but not similar names, near duplicates in other features, or provider-level labels and dimensions removed before splitting. Selecting the best-supported of 16 partitions, constructing four market and era clients, and intersecting modalities all condition the population, so results compare fitted rules on this benchmark rather than predicting outcomes for real organizations.

Statistical and Dataset Validity

Paired seeds capture partition, allocation, and training variance on one corpus, not independent corpora; recurring families and clients limit independence, and subgroups have fewer seeds. Leave-one-seed-out means ranged from 11.06 to 12.65 CTK points, all above the 3-point threshold, which complements rather than replaces the BCa interval. Only the three prespecified contrasts are Holm adjusted. Worst-client estimates depend on which client is worst in each arm and say little about a new organization. LAMDA's collection and labeling, together with market filtering, do not represent malware currently in the wild, and benign labels rest on zero antivirus detections. Different labeling could change the families, and external testing was not possible because McNdroid overlaps the dataset. Fingerprints confirm which data were used, not that they are representative.

External and Operational Validity

The study is offline: client dropout, communication cost, energy, latency, analyst workload, and on-device deployment were not evaluated, nor were privacy leakage, poisoning, evasion, or secure aggregation. Deployment would require evidence on applicability, cost, and security.

Conclusion

Complementary peer exposure increased FedAvg federation-wide unseen-family recall by 11.96 points relative to absence everywhere (95% BCa CI 9.15–15.21). The matched-budget three-arm design attributes this gain to complementary exposure, while the residual pooling effect remains uncertain, a distinction a local-versus-federated comparison cannot make.

The gain varies by family and has costs. FedAvg improved own-domain unseen-family recall by 3.26 points over local learning but lost 3.61 points on nonhidden malware and reduced AUROC and average precision; some families improved beyond local, some only recovered pooling losses, and some stayed hard. Permutation and wrong-family controls tie the gain to correct-family exposure, and exact dosing shows that it grows with exposure without a common saturation point. These results come from an offline benchmark with constructed clients and per-client calibration, and they indicate where federated learning can add value over local learning rather than guaranteeing it.

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Key Terms and Definitions

Complementary Threat Knowledge: The detection benefit associated with peer exposure to selected malware families, measured here by comparing matched collaborative conditions with and without those families.

Federated Learning: Joint model training in which participants fit models on their respective data and combine updates through an aggregation process.

Android Malware Family: A supplied grouping of malicious Android applications treated as related for the purposes of exposure selection and evaluation.

Locally Unseen Family: A selected malware family absent from a target client's fit sample, regardless of whether its examples occur at peers or share features with other malware.

Controlled Exposure: An experimental manipulation of which clients' training samples contain selected family examples while preserving specified comparison conditions.

Residual Collaboration Effect (Pooling): The recall difference between collaboration without the selected hidden families anywhere and the corresponding local learner, including differences in training composition and optimization schedules.

Negative Transfer: A reduction in a specified target performance measure after a defined transfer or collaboration intervention, relative to its stated reference.

Realized False-Positive Rate: The proportion of benign test applications flagged as malware by the calibrated predictor, distinct from the nominal calibration target.

Effective Peer Dose: The number of selected hidden-family examples actually used to fit peer models after exclusions, availability limits, and training budgets are applied.

Family-Macro Recall: An average that gives each eligible evaluated target-family entry equal weight, rather than weighting it by its number of test applications.